

Annie Prasanna Manoharan

+1(479)301-1366 | anniepr0501@gmail.com | <https://annie0159.github.io/Portfolio/> | <https://github.com/Annie0159>

EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Cryptography, Privacy Enhancing Techniques

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++, C, C#

Machine Learning & AI: PyTorch, Scikit-learn, NumPy, Pandas, Machine Learning, Deep Learning, Neural Networks, Computer Vision, Object Detection, Object Tracking, Image Processing, Predictive Modeling, Model Training, Model Validation

Generative AI: LLMs, Generative AI, Retrieval-Augmented Generation (RAG), RAG Pipelines, LLM Integration, Vector Embeddings, Semantic Search, LangChain, FAISS, Ollama

Data Engineering: ETL, ELT, Data Pipelines, Data Ingestion, Data Extraction, Data Transformation, Data Validation, Data Cleaning, Data Processing, Batch Processing, Real-Time Data Processing, Workflow Automation

Databases: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, ClickHouse, Relational Databases, NoSQL, Database Design,

Cloud & Data Platforms: AWS, Amazon S3, Amazon EC2, Amazon RDS, Apache Airflow, Docker, Linux

Tools & Methodologies: Git, GitHub, CI/CD, Agile, Scrum, Jira, Grafana, Streamlit

PROJECTS

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, SQL, Python, ClickHouse, Grafana

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to dynamically compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to cleanly handle out-of-order data streams
- Designed high-performance ClickHouse schemas using the MergeTree engine sorted by composite keys to physically align disk layouts with clinical querying patterns, while isolating ingestion via an optimized Python multi-topic batch consumer
- Integrated a Grafana visualization layer directly on top of the ClickHouse OLAP tier, delivering real-time clinical dashboards that monitor raw ventilator telemetry and instantly surface streaming Z-score anomalies

Local LLM-Powered News Research Platform

Python, LangChain, Streamlit, FAISS, Ollama

- Developed an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, and process unstructured web content on-demand
- Designed a local vector storage architecture using FAISS and nomic-embed-text to serialize high-dimensional embeddings, bypassing redundant ingestion steps and reducing processing latency
- Integrated a local LLM retrieval system via LangChain to orchestrate semantic queries against the vectorized data using local Llama 3.2 instances

PROFESSIONAL EXPERIENCE

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

Project: [MiGa: Multi-Chicken Gait Assessment](#)

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking (MOT), and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a Multi-Bird Detection split of 4,569 frames with 11,422 bounding boxes and a Gait-Score Regression split spanning 55 expert-annotated sequences
- Implemented spatial normalization to map keypoints into local coordinate spaces to eliminate translational variance, achieving a keypoint mAP of 91.83% at 1,250 FPS alongside 0.93 MOTA and 0.91 IDF1 tracking accuracy

Project: [DualFit: Two-Stage Virtual Try-On via Warping and Synthesis](#)

- Developed a PyTorch-based two-stage Virtual Try-On (VTON) model combining flow-based garment warping with Res-UNet image synthesis, with preprocessing and inpainting pipelines to improve garment alignment and visual realism
- Trained and evaluated the model on the VITON-HD dataset, achieving 24.9 PSNR, 0.913 SSIM, and 5.261 FID

Project Engineer, Soliton Technologies, India

July 2022 – June 2024

- Developed a C# WPF application integrating Python and LabVIEW test workflows to automate testing and improve execution efficiency.
- Served as the sole developer for a customized Medtronic solution, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements.
- Implemented reusable automation and validation workflows, collaborating with cross-functional teams to debug system issues, optimize processes, and improve test coverage.